



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 30/7/2026

In addition to the part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : CHEM F242

Course Title : Chemical Experimentation I

Instructor-in-Charge : Dalip Kumar

Tutorial/Practical Instructors: Nitika Grover, Bharti Khungar, Paritosh Shukla,
Indresh Kumar, Partha Sarathi Addy, Dalip Kumar

Student Instructors: Reena Jangir, Nitika, Pragati Shrivastava, Sandip Kumar Deb, Shikha Soni, Tarun Jangir, Mainika Verma, Pratiksha Jangir, Harshit, Huma, Asim Mohammed K, Ankit Yadav, Dhruvjayoti Paul

- 1. Course Description:** The chemical experimentation course is based on laboratory experiments in organic chemistry, and the course is available for second-year M.Sc. Chemistry students.
- 2. Scope and Objective of the Course:** The course's main objective is to provide students with a firsthand experience of basic techniques used in an organic chemistry laboratory. Broadly, the experiments include qualitative organic analysis, detection of known and unknown functional groups, chemical separation of a mixture of two compounds, examples of single-step and two-step organic syntheses, extraction of organic compounds from natural sources or from drugs, chromatographic separation of the pure compound from a mixture obtained in a multistep reaction, etc.
- 3. Textbooks:** Vogel's Textbook of Practical Organic Chemistry, 5th Ed.; Lab manual will be provided
- 4. Reference Books:** Reference sources for each experiment will be specified as necessary.
- 5. Course Plan:** The students will perform several experiments in qualitative, synthetic organic chemistry, including single-step and multi-step synthesis, extraction of natural products, and training in chromatographic separation, with an emphasis on individual planning and execution of the experiments. The students will be divided into two sections (I & II), and each section will conduct a set of experiments (Lab I or Lab-II experiments) in the first half semester, and thereafter, interchanging of the students will proceed during the second half of the semester. Selected 20 experiments will be discussed and performed in the class.

Lab-I Experiments: Synthesis, Extraction, Estimation, and Purification Techniques		
Expt No	Experiment	Learning Outcomes
1	Chemoselective reduction of an organic compound	• Safe laboratory practices and handling of glassware, equipment, and reagents. • Findings of limiting reagent and percent yield of a single or a multi-step synthesis. • Understanding reaction monitoring and product purification techniques like distillation, recrystallization, aqueous extraction, thin layer, and column chromatography. • Utilization of characterization techniques like mp, IR, NMR, etc. • Understanding the mechanism of some simple organic reactions Proper documentation of laboratory work
2	Multicomponent reaction (Biginelli Reaction)	
3	Oxidation of an organic Compound (Benzil from benzoin)	
4	Synthesis of benzilic acid from benzil using Grinding Technique	
5	Nucleophilic Aromatic Substitution	
6	Separation/purification of an organic compound using column chromatography	
7	Fischer Indole synthesis	
8	Coumarin synthesis using Microwave Irradiation	
9	Isolation of Curcumin from Curcuma Longa (Turmeric)	
10	Photodimerization of Benzophenone using Photochemical reactor	



Lab-II Experiments*: Qualitative Analysis of Organic Compounds

Expt No	Experiment	Learning Outcomes
1	Elemental Analysis of Known Organic Compounds (4 Compounds, common)	• Safe laboratory practices and handling of glassware, equipment, and reagents. • Preliminary observation of general physical characteristics of organic compounds • Understanding of the organic compounds chemical separation and purification techniques like distillation, extraction, etc. • Learning of elements and functional group analysis. • Utilization of characterization techniques like mp, IR, and NMR • Proper documentation of laboratory work
2-3	Functional group analysis of known compounds (12 Compounds, common) Non-nitrogenous functional groups: Alcohol, Phenol, Aldehyde, Ketone, Alkene, Alkyne (unsaturated compounds), Carbohydrate, Carboxylic acid, and Ester Nitrogenous functional groups: Amine, Amide, Nitroaromatics	
4-5	Complete analysis (Elemental analysis & functional group determination) of unknown compounds, & mp determination (min. of two compounds by each student)	
6	Separation of the mixture by Separation Technique-I: A mixture of water-soluble components and the water-insoluble component will be separated by water and identification of residue by elemental & functional group analysis (one mixture to each student)	
7-8	Separation of the mixture by Separation Technique-II: separation using NaHCO_3 & Technique-III: separation using NaOH . A mixture of acidic/basic and neutral components will be separated, and identification of acidic/basic and neutral components will be by elemental and functional group analysis (two different types of mixtures to each student)	
9	Separation of the mixture by Separation Technique-IV: separation using HCl ; A mixture of acid and base components will be separated, and their analysis (elemental and functional group) will be performed (one mixture to each student)	
10	Separation of the mixture by Separation Technique-V: A mixture of solid/liquid or liquid/liquid will be separated and a detailed analysis of both the components will be performed via elemental and functional group analysis (one mixture to each student)	

*In a qualitative analysis of compounds/mixtures, NMR and IR will be provided to characterize the compounds.

6. Evaluation Scheme:

Component	Weightage (%)	Date & Time
Laboratory Work & Reports	84 % (252 Marks)	Continuous
Lab tests I and II	16 % Test-I & Test-II (48 Marks)	To be announced

7. **Marks Distribution:** Each Lab experiment will be of 14 Marks. The distribution of marks is as follows:

Lab-Experiments: Attendance (2); Lab quiz (3); Performance and Other evaluation components (9).

8. **Notices:** All notices concerning the course will be displayed on Nalanda.

9. **Make-up Policy:** Since there are TWO buffers, hence NO MAKEUP will be available in this course for whatsoever reasons.

10. **Minimum performance:** To clear the course, a student should obtain 30% of the average marks of the top 10 performers in the class, or 40% of the median (whichever is lower).

11. **Attendance Policy:** For continuous evaluation linked to attendance, final grades will be awarded only to students who maintain a minimum attendance of 50% (two marks have been assigned for being participation and punctual in each class).

Instructor-in-charge
CHEM F242